

Boss Challenge — Paper 02 (Class 7)

Light | Nutrition in Animals | Fractions & Decimals

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. Light normally travels in:
(A) curved paths
(B) straight lines
(C) a zig-zag
(D) circles
2. Chewing a piece of bread for a long time makes it taste sweet because saliva:
(A) adds sugar
(B) starts breaking starch into sugar
(C) cools the bread
(D) kills germs
3. Find: $\frac{1}{2} \times \frac{3}{4}$
(A) $\frac{3}{8}$
(B) $\frac{4}{6}$
(C) $\frac{3}{2}$
(D) $\frac{1}{8}$
4. The image in a plane (flat) mirror is:
(A) real and inverted
(B) virtual, erect and the same size
(C) smaller and real
(D) larger than the object
5. The largest gland in the human body, which makes bile, is the:
(A) stomach
(B) liver
(C) kidney
(D) heart
6. What is $\frac{1}{3}$ of 90?
(A) 30
(B) 27
(C) 33
(D) 60
7. 'AMBULANCE' is written reversed on the front of the van so a driver ahead reads it correctly in the mirror. This happens because a plane mirror causes:
(A) magnification
(B) lateral inversion (left-right swap)
(C) dispersion
(D) no change

8. The finger-like projections in the small intestine that absorb digested food are:

- (A) villi
- (B) teeth
- (C) muscles
- (D) glands

9. The reciprocal (multiplicative inverse) of $\frac{2}{5}$ is:

- (A) $\frac{5}{2}$
- (B) $\frac{2}{5}$
- (C) $-\frac{2}{5}$
- (D) 10

10. The law of reflection says the angle of incidence is:

- (A) always 90°
- (B) equal to the angle of reflection
- (C) always larger than the reflection angle
- (D) zero

11. The stomach releases acid and juices mainly to:

- (A) cool the food
- (B) digest food and kill many germs
- (C) add colour
- (D) make the food sweet

12. Find: $\frac{3}{4} \div \frac{1}{2}$

- (A) $\frac{3}{8}$
- (B) $\frac{3}{2}$
- (C) $\frac{1}{2}$
- (D) $\frac{2}{3}$

13. A concave mirror is used in a torch/headlight because it can:

- (A) spread light everywhere
- (B) throw the light into a strong parallel beam
- (C) split light into colours
- (D) make the bulb cooler

14. A cow brings food back from its stomach to chew again. This is called:

- (A) digestion
- (B) chewing the cud (rumination)
- (C) absorption
- (D) egestion

15. Find: 0.2×0.3

- (A) 0.6
- (B) 0.06
- (C) 0.5
- (D) 6.0

16. An Amoeba captures its food using:

- (A) teeth
- (B) false feet (pseudopodia)
- (C) a tongue
- (D) roots

17. A convex mirror is used as a vehicle's rear-view mirror because it gives:

- (A) a magnified inverted image
- (B) an erect, smaller image and a wider view
- (C) colours
- (D) no image

18. Find: $2.5 \div 0.5$

- (A) 0.5
- (B) 5
- (C) 1.25
- (D) 2

19. An image that can be caught on a screen is called a:

- (A) virtual image
- (B) real image
- (C) mirror image
- (D) shadow

20. Teeth used mainly for cutting food are the:

- (A) incisors
- (B) canines
- (C) molars
- (D) premolars

21. Which is greater: $\frac{2}{3}$ or $\frac{3}{4}$?

- (A) $\frac{2}{3}$
- (B) $\frac{3}{4}$
- (C) they are equal
- (D) cannot say

22. White sunlight passing through a glass prism splits into seven colours. This splitting is called:

- (A) reflection
- (B) dispersion
- (C) absorption
- (D) conduction

23. Most absorption of digested food into the blood happens in the:

- (A) mouth
- (B) stomach
- (C) small intestine
- (D) large intestine

24. Find: $\frac{1}{2} + \frac{1}{3}$

- (A) $\frac{2}{5}$
- (B) $\frac{5}{6}$
- (C) $\frac{1}{6}$
- (D) $\frac{2}{6}$

25. The seven colours of white light in order are remembered as VIBGYOR. Which is NOT one of them?

- (A) violet
- (B) indigo
- (C) black
- (D) orange

26. The large intestine mainly:

- (A) digests proteins
- (B) absorbs water from the undigested food
- (C) makes bile
- (D) cuts food

27. 0.75 written as a fraction (in lowest terms) is:

- (A) $\frac{3}{4}$
- (B) $\frac{7}{5}$
- (C) 75
- (D) $\frac{1}{4}$

28. A convex lens (thick in the middle) is commonly used as a:

- (A) magnifying glass
- (B) plane mirror
- (C) prism
- (D) torch reflector

29. Bile, which helps digest fats, is made by the:

- (A) stomach
- (B) liver
- (C) tongue
- (D) teeth

30. Find: $\frac{2}{5}$ of 100

- (A) 20
- (B) 40
- (C) 25
- (D) 50

31. A concave lens (thin in the middle):

- (A) converges light
- (B) spreads (diverges) light
- (C) makes colours
- (D) reflects light

32. Cud-chewing animals can digest grass (cellulose) because they have helpful ____ in their stomach.

- (A) acids only
- (B) bacteria
- (C) teeth
- (D) saliva

33. Find: 1.2×5

- (A) 0.6
- (B) 6
- (C) 60
- (D) 1.7

34. We can see a non-luminous object (like a book) because it:

- (A) makes its own light
- (B) reflects light into our eyes
- (C) absorbs all light
- (D) is transparent

35. The four kinds of teeth in humans are incisors, canines, premolars and molars. Wisdom teeth are a type of:

- (A) incisor
- (B) canine
- (C) molar
- (D) premolar

36. Find: $\frac{5}{6} - \frac{1}{3}$

- (A) $\frac{4}{3}$
- (B) $\frac{1}{2}$
- (C) $\frac{6}{3}$
- (D) $\frac{1}{6}$

37. The number of images of an object placed between two parallel plane mirrors is:

- (A) one
- (B) two
- (C) infinite (very many)
- (D) none

38. The wave-like squeezing of the food-pipe walls that pushes food down is called:

- (A) peristalsis
- (B) digestion
- (C) absorption
- (D) reflection

39. Find: $7/10$ as a decimal

- (A) 0.07
- (B) 0.7
- (C) 7.0
- (D) 1.7

40. In a plane mirror, the image is formed:

- (A) on the mirror's surface
- (B) as far behind the mirror as the object is in front
- (C) very far behind always
- (D) in front of the mirror

41. If a person's intestinal villi are damaged, what will be affected the most?

- (A) seeing
- (B) absorption of digested food
- (C) hearing
- (D) breathing

42. Which decimal lies between 0.3 and 0.4?

- (A) 0.25
- (B) 0.35
- (C) 0.45
- (D) 0.5

43. A periscope (to see over a wall or from a submarine) works using:

- (A) two plane mirrors
- (B) a prism only
- (C) a magnet
- (D) a battery

44. A cow's stomach has a special large part called the:

- (A) rumen
- (B) villus
- (C) molar
- (D) liver

45. Find: $0.6 + 0.45$

- (A) 1.05
- (B) 0.51
- (C) 0.105
- (D) 1.5

46. The colours of white light combine back to white. A spinning disc that shows this is called:

- (A) Newton's colour disc
- (B) a prism
- (C) a periscope
- (D) a kaleidoscope

47. Saliva is produced by glands found in the:

- (A) stomach
- (B) mouth
- (C) small intestine
- (D) liver

48. Find: $\frac{3}{5} \times 10$

- (A) 2
- (B) 6
- (C) 30
- (D) 15

49. A red ball looks red because it:

- (A) makes red light
- (B) reflects red light and absorbs the other colours
- (C) absorbs all light
- (D) is transparent to red

50. Villi are useful because they:

- (A) make the intestine longer only
- (B) greatly increase the surface area for absorption
- (C) produce acid
- (D) chew food

51. Find: $1 \div \frac{1}{4}$

- (A) 4
- (B) $\frac{1}{4}$
- (C) 1
- (D) 0.25

52. You cannot see around a corner because light:

- (A) travels in straight lines
- (B) travels in curves
- (C) is too fast
- (D) is invisible

53. Which of these is NOT a step of nutrition in animals?

- (A) ingestion
- (B) digestion
- (C) reflection
- (D) absorption

54. Find: $9/10 - 0.4$

- (A) 0.5
- (B) 0.6
- (C) 0.05
- (D) 1.3

55. A dentist uses a small concave mirror to look at teeth because it gives:

- (A) a smaller image
- (B) an enlarged (magnified) image when held close
- (C) colours
- (D) no image

56. Permanent teeth replace the:

- (A) wisdom teeth
- (B) milk (temporary) teeth
- (C) molars only
- (D) canines only

57. Find: half of $2/3$

- (A) $1/3$
- (B) $2/6$ only
- (C) $4/3$
- (D) $1/6$

58. A torch's shiny curved reflector behind the bulb is shaped like a:

- (A) convex mirror
- (B) concave mirror
- (C) flat mirror
- (D) prism

59. The acid present in the human stomach is mainly:

- (A) sulphuric acid
- (B) hydrochloric acid
- (C) nitric acid
- (D) citric acid

60. Find: 0.05×100

- (A) 0.5
- (B) 5
- (C) 500
- (D) 50

61. Which mirror can form an image LARGER than the object (when the object is close)?

- (A) plane mirror
- (B) concave mirror
- (C) convex mirror
- (D) none

62. The opening that leads to the food pipe is close to the one leading to the windpipe, so we should:
- (A) talk a lot while eating
 - (B) not talk or laugh while eating, to avoid choking
 - (C) eat very fast
 - (D) skip chewing
63. Find: $(2/3) \div (4/9)$
- (A) $8/27$
 - (B) $3/2$
 - (C) $6/12$ only
 - (D) $2/3$
64. The image formed by a plane mirror has a magnification of:
- (A) 1 (same size)
 - (B) 2 (double)
 - (C) $1/2$ (half)
 - (D) 0
65. The tongue helps in eating by:
- (A) only swallowing
 - (B) mixing food with saliva and tasting it
 - (C) making bile
 - (D) absorbing food
66. Find: $0.2 + 0.2 + 0.2 + 0.2 + 0.2$
- (A) 1.0
 - (B) 0.10
 - (C) 2.0
 - (D) 0.5
67. A kaleidoscope makes beautiful repeating patterns using:
- (A) mirrors set at an angle
 - (B) a battery
 - (C) a magnet
 - (D) a lens only
68. Cellulose in grass is digested by cows but NOT by humans because humans:
- (A) have no teeth
 - (B) lack the helpful bacteria that break down cellulose
 - (C) don't eat grass
 - (D) have a liver
69. Find: $1 - 3/8$
- (A) $5/8$
 - (B) $2/8$
 - (C) $3/8$
 - (D) $1/8$

70. Sunlight (white light) is actually made of:

- (A) only white
- (B) seven colours mixed together
- (C) only red
- (D) no colour

71. Water is mainly absorbed from the undigested food in the:

- (A) mouth
- (B) stomach
- (C) small intestine
- (D) large intestine

72. Find: $4.5 \div 1.5$

- (A) 3
- (B) 0.3
- (C) 30
- (D) 2

73. A shadow forms behind an object because light:

- (A) bends around it
- (B) travels in straight lines and is blocked
- (C) is absorbed fully
- (D) changes colour

74. Premolars and molars are mainly used to:

- (A) cut food
- (B) tear food
- (C) grind and chew food
- (D) taste food

75. Find: $\frac{2}{3} \times 9$

- (A) 3
- (B) 6
- (C) 18
- (D) 12

76. Light from the Sun reaches the Earth. The Moon is seen because it:

- (A) makes its own light
- (B) reflects sunlight
- (C) absorbs all light
- (D) is a star

77. The four types of human teeth are:

- (A) incisors, canines, premolars, molars
- (B) only molars
- (C) only incisors
- (D) fangs and claws

78. Find: 0.1×0.1

- (A) 0.1
- (B) 0.01
- (C) 1.0
- (D) 0.2

79. Two mirrors face each other in a barber's shop, showing many images of you. This is because:

- (A) the mirrors are coloured
- (B) each mirror keeps re-reflecting the other's image
- (C) of magnets
- (D) of electricity

80. Egestion means:

- (A) taking in food
- (B) removing the undigested waste from the body
- (C) absorbing food
- (D) chewing

81. Find: $\frac{3}{4}$ of 20

- (A) 5
- (B) 10
- (C) 15
- (D) 16

82. To send light far in a straight beam (searchlight), the best reflector is:

- (A) plane
- (B) concave
- (C) convex
- (D) none

83. A child who cannot absorb nutrients well even though they eat enough may have a problem with their:

- (A) ears
- (B) intestinal villi
- (C) nose
- (D) nails

84. Find: 0.25 of 200

- (A) 25
- (B) 50
- (C) 75
- (D) 100

85. A pencil dipped in a glass of water looks bent at the surface. This happens because light:

- (A) reflects
- (B) bends (refracts) when passing from water to air
- (C) disappears
- (D) changes colour

86. Most chemical digestion is completed in the:

- (A) mouth
- (B) stomach
- (C) small intestine
- (D) large intestine

87. Find: $\frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

- (A) $\frac{3}{4}$
- (B) $\frac{1}{12}$
- (C) $\frac{3}{12}$ only
- (D) 1

88. Which is a luminous object (makes its own light)?

- (A) the Moon
- (B) a mirror
- (C) the Sun
- (D) a book

89. Saliva acts on which food the most?

- (A) fats
- (B) starch (carbohydrate)
- (C) proteins
- (D) minerals

90. A ribbon $\frac{3}{4}$ m long is cut into pieces of $\frac{1}{4}$ m each. How many pieces?

- (A) 2
- (B) 3
- (C) 4
- (D) 6

91. When you look into a spoon's shiny back (bulging side), your image is small and erect. The spoon's back acts like a:

- (A) plane mirror
- (B) convex mirror
- (C) concave mirror
- (D) prism

92. Which animal is a ruminant (chews the cud)?

- (A) lion
- (B) cow
- (C) amoeba
- (D) frog

93. Find: $\frac{2}{3} + \frac{1}{6}$

- (A) $\frac{3}{9}$
- (B) $\frac{5}{6}$
- (C) $\frac{1}{2}$
- (D) 1

94. A magnifying glass makes small print look bigger. It is a:

- (A) concave lens
- (B) convex lens
- (C) plane mirror
- (D) prism

95. Assimilation in nutrition means the body:

- (A) throws out waste
- (B) uses the absorbed food to build and repair the body
- (C) chews food
- (D) tastes food

96. Find: $\frac{5}{8} - \frac{1}{4}$

- (A) $\frac{3}{8}$
- (B) $\frac{4}{4}$
- (C) $\frac{1}{2}$
- (D) $\frac{6}{8}$

97. Which surface reflects almost all the light falling on it, looking bright white?

- (A) a black cloth
- (B) a clean mirror or white sheet
- (C) a dark wall
- (D) a hole

98. Which is the correct order food travels?

- (A) mouth -> stomach -> small intestine -> large intestine
- (B) stomach -> mouth -> large intestine
- (C) mouth -> large intestine -> stomach
- (D) small intestine -> mouth -> stomach

99. Find: $\frac{7}{10} + 0.3$

- (A) 1.0
- (B) 0.10
- (C) 0.73
- (D) 1.3

100. A convex lens used in the right way can form a real, inverted image on a screen. A plane mirror, in contrast, always forms an image that is:

- (A)** real and inverted
- (B)** virtual and erect
- (C)** on a screen
- (D)** upside down